The Encyclopedia of AI-Assisted Software Development

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Building world-class software with AI is a vocabulary problem before it is a coding problem: the AI writes whatever you can specify precisely, so this encyclopedia teaches the method first and then the vocabulary that makes the method precise.

Contents

- How to use this encyclopedia
- Part I — Core properties
- Part II — Architecture and scaling
- Part III — Data and databases
- Part IV — The method: prompting AI for industry-grade results
- Part V — The build loop and project rules
- Part VI — APIs, networking and communication
- Part VII — Frontend architecture
- Part VIII — UI and UX design
- Part IX — Visual and special effects
- Part X — Reliability and resilience
- Part XI — Performance, optimization and cost
- Part XII — Testing and verification
- Part XIII — Security and protecting your work
- Part XIV — Observability and operations
- Part XV — Distributed systems
- Part XVI — Pipelines, CI/CD and infrastructure
- Part XVII — Engineering process and product thinking
- Part XVIII — Domain notes: trading, ML and LLM systems
- Part XIX — The system design interview
- Part XX — Study path and prompt templates
- Part XXI — Concurrency, boundaries and safe change
- Part XXII — Practices and common subsystems
- Part XXIII — APIs from zero to hero
- Part XXIV — One system, designed end to end

- Verification note

How to use this encyclopedia

The parts are ordered to match the study path in Part XX, so reading straight through is the intended route, and each part afterwards works as a dictionary you consult while building.

The sequence is deliberate: core vocabulary first, then the method for directing an AI, then building, hardening, scaling, and the professional practices that surround the work. Parts IV and V teach how to get industry-grade output from an AI and sit after the foundations rather than before them, because you cannot ask for idempotent writes until you know what idempotency is. Each entry gives the term, a plain-language meaning, and where useful a phrase you can paste into a prompt — that third column is how vocabulary turns into working software.

If you want to	Go to
Get better output from AI tomorrow	Parts IV and V, then Part XX
Learn the core vocabulary	Parts I to III
Build and ship something usable	Parts IV to IX
Harden what you already built	Parts X to XIV
Scale it, or survive failure	Parts XV and XVI
Prepare for system design interviews	Parts I to III and VI, then Part XIX
Work like a team, not a solo hacker	Part XVII

Three rules that make the rest work. First, a term you cannot explain in one sentence is a term you cannot prompt with — read the meaning column, not just the term. Second, breadth without depth is worthless: after reading, pick one project and implement five of these concepts properly rather than fifty badly. Third, verify everything the AI claims; the single largest quality gap between amateur and industry-grade AI-assisted work is whether the output was checked or merely accepted.

Part I — Core properties

These seven words describe what a system is, and nearly every design argument reduces to trading one of them against another.

Term	Meaning	Example
Idempotency	Repeating a request leaves the same result as doing it once	A payment call with an idempotency key cannot double-charge on retry
Latency	Time for one request to complete	120 ms API response
Throughput	Requests handled per unit of time	10,000 requests per second
Availability	Share of time the system is usable	99.99% is about 52 minutes of downtime a year
Reliability	Works correctly over time without losing or corrupting data	No lost orders after a crash
Scalability	Handles growing load without redesign	Adding servers absorbs 10x traffic
Fault tolerance	Keeps working when parts fail	One database node dies, service continues
Durability	Committed data survives crashes	Write is on disk before acknowledging
Consistency	All readers see the same data	Every client sees the same balance
Elasticity	Capacity grows and shrinks with demand	Autoscaling down overnight
Maintainability	Cost of changing the system later	A new engineer ships in week one

Why idempotency comes first

Networks fail and clients retry. If a write is not idempotent, a retry creates a duplicate — two orders, two charges, two rows. The standard fix: the client sends a unique idempotency key with the request; the server stores the outcome against that key and returns the stored result for any repeat. Make every state-changing endpoint idempotent and an entire category of production incidents disappears.

A useful check: `DELETE /order/9` is idempotent, because the order is absent after the first call and every later call leaves that same state. `POST /orders` is not, unless you add a key.

The tradeoffs that recur

Tension	What you give up
Latency vs throughput	Batching raises throughput and raises latency
Consistency vs availability	Under a network partition you can have one, not both
Consistency vs latency	Waiting for all replicas to agree costs milliseconds
Simplicity vs flexibility	Every abstraction added is a cost paid forever
Speed of delivery vs maintainability	Shortcuts are borrowed time at interest

Part II — Architecture and scaling

Architecture is the set of decisions that are expensive to reverse, so the goal is to defer what you can and get the boundaries right.

Scaling

Term	Meaning
Vertical scaling	A bigger machine: more CPU, RAM, disk. Simple, with a hard ceiling
Horizontal scaling	More machines behind a load balancer. Harder, but open-ended
Load balancer	Distributes traffic across servers; also removes failed ones from rotation
Stateless service	Keeps no session data between requests, so any instance can serve any request
Sticky session	Routes a user to the same instance; a sign state has leaked into the service
Sharding	Splitting data across databases by a key, such as user ID ranges
Hot shard	One partition receiving disproportionate traffic; the usual sharding failure
Replication	Copying data to other nodes for reads and redundancy

Term	Meaning
Primary and replica	Writes go to one node, reads spread across copies
Connection pooling	Reusing database connections instead of opening one per request
Autoscaling	Adding and removing instances based on load metrics

Structure and boundaries

Term	Meaning	Prompt phrase
Separation of concerns	Each module does one job: routes, logic, data access	"Keep routes, services and data access in separate layers"
Coupling and cohesion	Low coupling between modules, high cohesion within one	"Modules communicate only through interfaces"
Monolith	One deployable unit; simplest and correct for most projects	"Modular monolith, no microservices"
Modular monolith	One deployment, strict internal module boundaries	The right default for solo and small-team work
Microservices	Independently deployable services; organisational tool, not a performance one	Justified by team size, rarely by load
Distributed monolith	Microservices that must deploy together: all the cost, none of the benefit	The failure mode to avoid
Clean / hexagonal architecture	Business logic at the centre, I/O at the edges behind ports	"Business logic must not import the database client"
Dependency injection	Dependencies passed in rather than constructed inside	"Inject the client so I can mock it"
Abstraction layer / adapter	A wrapper hiding a vendor or external API	"Wrap the broker API behind an adapter"
Single source of truth	One owner per piece of data; others read from it	"State lives in one place only"

Term	Meaning	Prompt phrase
Twelve-factor app	Config in environment, stateless processes, disposable instances	A checklist worth reading once

Common patterns

Pattern	Use when
API gateway	One entry point handling auth, routing and rate limiting
Backend for frontend (BFF)	Web and mobile need differently shaped responses
Sidecar	Cross-cutting concerns (logging, proxying) run beside the service
Strangler fig	Replacing a legacy system incrementally rather than rewriting
Bulkhead	Isolating resource pools so one overload cannot sink everything
Cache-aside	Read the cache, fall back to the database, then populate the cache

Part III — Data and databases

The database is where most correctness bugs and most performance problems live, so it repays study more than any other layer.

Transactions and guarantees

Term	Meaning
ACID	Atomicity, Consistency, Isolation, Durability: the transaction contract
Transaction	A group of operations that all succeed or all roll back

Term	Meaning
Isolation levels	Read uncommitted, read committed, repeatable read, serializable — weaker is faster and buggier
Dirty read	Reading another transaction's uncommitted change
Non-repeatable read	The same row returns different values within one transaction
Phantom read	New rows appear in a repeated range query
MVCC	Multi-version concurrency control: readers see a snapshot and never block writers
Deadlock	Two transactions each waiting on the other's lock
Optimistic locking	Version column checked at write time; retry on conflict
Pessimistic locking	Lock the row up front; safer, less concurrent
CAP theorem	Under a network partition, choose consistency or availability
PACELC	Extends CAP: even without a partition, there is a latency-versus-consistency choice
Eventual consistency	Replicas converge over time; a like count can lag
Strong consistency	Every read sees the latest committed write
Read-your-writes	A user always sees their own change, even under eventual consistency

Storage internals

Term	Meaning
B-tree index	Balanced tree; fast reads and range scans. Default in PostgreSQL and MySQL
LSM tree	Log-structured merge tree; write-optimised. Used by Cassandra, RocksDB
WAL	Write-ahead log: changes are logged before being applied, which is how crash recovery works

Term	Meaning
Clustered index	Table rows physically ordered by the index key
Covering index	Index contains every column the query needs, so the table is never touched
Composite index	Multi-column index; column order determines which queries it serves
Cardinality	Number of distinct values; low cardinality indexes are nearly useless
Query planner	Decides how to execute a query; `EXPLAIN ANALYZE` shows its choice
Full table scan	Reading every row: the usual cause of a slow query
N+1 query	One query per row in a loop instead of one join; the classic ORM trap
Vacuum / compaction	Reclaiming space from deleted or superseded rows

Modelling and operations

Term	Meaning
Normalization	Removing duplication so each fact is stored once
Denormalization	Deliberate duplication to make reads faster
Foreign key	Enforced reference between tables
Migration	A versioned, repeatable schema change; never edit production schema by hand
Backfill	Populating a new column or table for existing rows
Soft delete	Marking rows deleted rather than removing them
Partitioning	Splitting one table by range or hash within a database
OLTP vs OLAP	Transaction workloads (many small writes) versus analytical ones (large scans)
Columnar store	Stores by column, ideal for analytics. DuckDB, ClickHouse, Parquet
Time-series database	Optimised for timestamped append-heavy data

Term	Meaning
Data warehouse / lake	Central analytical store; the lake keeps raw files

Choosing a store

Need	Choose
Relational integrity, transactions	PostgreSQL
Key-value cache, ephemeral state	Redis
Flexible documents, early-stage schema	MongoDB
Analytical queries over local files	DuckDB
Huge write volume, tunable consistency	Cassandra
Full-text or fuzzy search	Elasticsearch, OpenSearch
Vector similarity for AI retrieval	pgvector, Qdrant

Default to PostgreSQL until you have a measured reason not to; most "we need NoSQL" conclusions are really missing indexes.

Part IV — The method: prompting AI for industry-grade results

Stop prompting for code and start prompting for engineering process: the quality of AI output is set by the constraints you impose and the verification you demand, not by clever wording.

The core techniques

Technique	Weak prompt	World-class prompt
Role and quality bar	"Build a trading bot"	"Act as a principal engineer at a production trading firm. Optimize for correctness and auditability over speed."
Spec first	Jumps straight to code	"Write the spec first: goals, non-goals, schemas, failure modes. Wait for my approval before coding."
Plan before build	One giant build request	"Give a milestone plan with acceptance criteria per step. Build only step 1."

Technique	Weak prompt	World-class prompt
Demand tradeoffs	Accepts the first answer	"Give 3 architectures with tradeoffs. Recommend one and say under what conditions you'd be wrong."
Hard constraints	Vague stack and limits	"Python 3.12, FastAPI, no new dependencies, idempotent writes, p99 under 200ms."
Test-first	Tests as an afterthought	"Write failing tests first, then the minimal code to pass them."
Adversarial review	"Looks good?"	"Act as a hostile reviewer. Find race conditions, data-loss paths and security holes. Rank by severity."
Pre-mortem	Hopes for the best	"Assume this failed in production in 3 months. List the 10 likeliest causes and their mitigations."
Failure injection	Happy path only	"Write chaos tests: network drop, duplicate messages, partial writes, clock skew."
Evidence rule	Trusts claims	"Never claim it works without running it. Show command output. Mark anything unverified as UNVERIFIED."
Small diffs	Giant rewrites	"Change only these files. Show the diff. Do not refactor anything else."
Explain-back	Pastes and moves on	"Explain this design as if teaching a junior, then list its 3 weakest points."
Persistent rules	Repeats itself each session	Keep a project rules file (Part II) that every session loads automatically.

Questions worth asking every session

- "What are you assuming that I haven't told you? Ask before building."
- "What would make this design fail at 100x load?"
- "Which parts of this are you least confident about?"
- "Delete everything non-essential. What is the simplest version that meets the spec?"
- "What did you change that I did not ask for?"

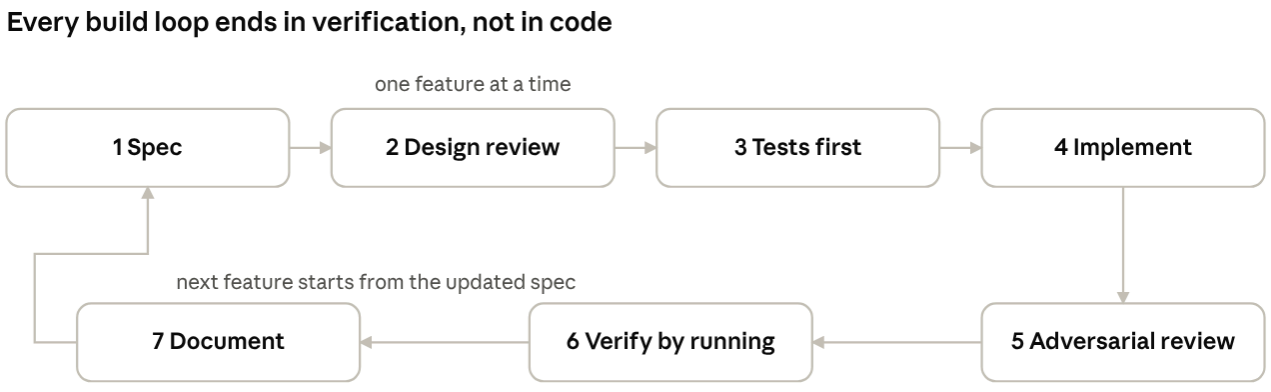
Anti-patterns that produce amateur output

Anti-pattern	Why it fails	Instead
One mega-prompt	Too many degrees of freedom; the model guesses	One feature per session, one commit per step
Accepting unread code	You cannot debug or defend what you did not read	Read every diff; ask for an explanation if unclear
Vague quality words ("make it good")	Unmeasurable, so unenforceable	Name the measurable bar: latency, coverage, WCAG AA
No rejection criteria	Everything looks acceptable	State up front what would make you reject the output
Letting it refactor freely	Silent breakage outside the task	Scope the files it may touch
Trusting "it works"	Models assert more confidently than they verify	Demand command output and tests
Starting over on each bug	Loses working context	Fix forward in small, guarded steps

The mental model: you are the architect and reviewer; the AI is a fast, tireless engineer who needs a precise contract and will cheerfully build the wrong thing if the contract is loose.

Part V — The build loop and project rules

Industry-grade output comes from a repeating loop in which code is the fourth step, not the first.



the build loop · 7 steps, one feature per pass

Each pass handles one feature. Steps 5 and 6 are where amateur and professional work diverge: the adversarial review is run as a separate pass with a fresh perspective, and verification means running the thing, not reading it.

Step	What happens	Done when
1 Spec	You write goals, non-goals, schemas, constraints	The AI can restate it without asking questions
2 Design review	AI proposes options, you choose	Tradeoffs are written down, not just the choice
3 Tests first	Failing tests encode the contract	Tests fail for the right reason
4 Implement	Minimal code to pass, small diffs	Tests green, diff reviewed line by line
5 Adversarial review	Fresh pass hunting for flaws	Findings ranked by severity
6 Verify by running	Execute, measure, inspect output	You have command output, not assurances
7 Document	Decision record, README, runbook	A stranger could pick it up

The project rules file

A rules file (commonly `CLAUDE.md` at the repository root, or your IDE's equivalent) is loaded into every session so standards survive across conversations. It is the single highest-leverage artifact in AI-assisted development.

```
# Project rules

## Stack
Python 3.12, FastAPI, DuckDB, React + Vite. No new dependencies without
asking.

## Architecture
Layers: routes -> services -> repositories. Routes contain no business logic.
All external calls go through an adapter module.

## Non-negotiable
- Validate all input at the edge with Pydantic models.
- Writes are idempotent; external calls have timeouts and bounded retries.
- Secrets come from environment variables. Never hardcode or log them.
```

```
- Parameterised queries only.
- Structured JSON logs with a correlation ID.

## Testing
Write failing tests before implementation. Mock all external services.

## Working style
One feature per session. Show the diff. Change only files I name.
Never claim something works without showing the command output.
Mark anything unverified as UNVERIFIED.
Ask before refactoring anything outside the task.

## Never do
No placeholder or fabricated data. No silent exception swallowing.
No commented-out code left behind.
```

Keep it short enough to be read every session, update it whenever you correct the same mistake twice, and treat a rule you must repeat in chat as a rule missing from this file.

Part VI — APIs, networking and communication

Most debugging time is spent at boundaries, so knowing what happens between two machines is worth more than any framework.

What happens when a request is made

Layer	Term	Meaning
Naming	DNS	Resolves a hostname to an IP address; TTL controls caching
Transport	TCP	Ordered, reliable, connection-based. Three-way handshake
Transport	UDP	Fire-and-forget, no ordering. Used for video, games, DNS
Security	TLS	Encrypts the connection; handshake exchanges certificates and keys
Security	Certificate / CA	Proves server identity; expiry is a classic outage cause
Application	HTTP/1.1	One request at a time per connection; head-of-line blocking

Layer	Term	Meaning
Application	HTTP/2	Multiplexed streams over one connection
Application	HTTP/3 (QUIC)	Over UDP; faster setup, no transport-level head-of-line blocking
Routing	Reverse proxy	Sits in front of servers: TLS termination, routing, caching (Nginx)
Routing	Forward proxy / NAT	Sits in front of clients

API styles

Style	Strength	Weakness
REST	Simple, cacheable, universal tooling	Over- and under-fetching
GraphQL	Client asks for exactly the fields it needs	Caching and rate limiting are harder
gRPC	Binary, fast, typed contracts, streaming	Not browser-native, harder to debug by hand
Webhooks	Server pushes events to your URL	You must handle retries and duplicates

API design vocabulary

Term	Meaning	Prompt phrase
Resource and verb	Nouns in the path, actions as HTTP methods	"RESTful routes, no verbs in paths"
Idempotent methods	GET, PUT, DELETE are idempotent; POST is not	"Make POST /orders idempotent with a key"
Status codes	2xx success, 4xx client error, 5xx server error	"Correct status codes, no 200 with an error body"
Schema / contract	The agreed request and response shape	"Define Pydantic models first"
OpenAPI / Swagger	Machine-readable API description	"Generate an OpenAPI spec"
Validation	Reject malformed input at the edge	"Validate all input, fail fast"

Term	Meaning	Prompt phrase
Pagination	Offset-based (simple) or cursor-based (stable under writes)	"Cursor pagination, never return all rows"
Versioning	Change without breaking existing clients	"Prefix routes with /v1"
Backward compatibility	Add fields, never remove or repurpose them	"Additive changes only"
Rate limiting	Caps per client; token bucket is the usual algorithm	"60 requests per minute per key"
CORS	Browser rule controlling cross-origin calls	"Explicit origin allowlist, never a wildcard with credentials"
HATEOAS	Responses carry links to related actions	Rarely used in practice

Real-time transport

Option	Direction	Use when
Polling	Client asks repeatedly	Simplicity matters more than freshness
Long polling	Request held open until data exists	WebSocket is unavailable
Server-sent events (SSE)	Server to client only	Live feeds, notifications, token streaming
WebSocket	Both directions, persistent	Chat, live prices, collaborative editing

Asynchronous messaging

Term	Meaning
Message queue	Buffer decoupling producer from consumer (RabbitMQ, SQS)
Event streaming	Durable, replayable ordered log (Kafka, Redpanda)

Term	Meaning
Publish/subscribe	Publishers broadcast; many subscribers react independently
Fan-out	One event triggers many downstream actions
Dead letter queue	Holds messages that repeatedly fail, for inspection
Consumer lag	How far behind the consumer is; the key health metric

Part VII — Frontend architecture

The decisive frontend question is where state lives and where rendering happens; almost every other choice follows from those two.

State

Term	Meaning
Client state	State owned by the UI: open modals, form drafts, filters
Server state	A cache of data the server owns; needs fetching, caching, invalidation
State colocation	Keep state as close to where it is used as possible
Lifting state up	Move shared state to the nearest common ancestor
Prop drilling	Passing props through many layers; a sign state is in the wrong place
Global store	Redux, Zustand, Jotai for genuinely cross-cutting client state
Server-state library	TanStack Query, SWR: caching, revalidation, deduplication
Derived state	Computed from existing state; never stored separately
Optimistic update	Apply the change immediately, reconcile or roll back on response

Term	Meaning
Normalization	Store entities by ID rather than nested duplicates

The common mistake is putting server data in a global client store. Use a server-state library for anything fetched, and a store only for client state.

Rendering strategies

Strategy	How it works	Best for
CSR	Browser downloads JS and renders	Dashboards behind a login
SSR	Server renders HTML per request	Personalized, SEO-relevant pages
SSG	HTML built once at build time	Docs, blogs, marketing
ISR	Static pages regenerated on a schedule or on demand	Large catalogs
Streaming SSR	HTML sent in chunks as it becomes ready	Slow data without blocking the shell
Islands architecture	Static HTML with small interactive regions	Content sites with a little interactivity
Hydration	Attaching JS behaviour to server-rendered HTML	The cost SSR pays
RSC	Components rendered on the server, never shipped to the client	Reducing bundle size

Performance

Term	Meaning	Target
LCP	Largest contentful paint: when main content appears	Under 2.5s
INP	Interaction to next paint: responsiveness	Under 200ms
CLS	Cumulative layout shift: visual stability	Under 0.1
TTFB	Time to first byte	Under 800ms
Code splitting	Load JavaScript per route or feature	—

Term	Meaning	Target
Tree shaking	Eliminate unused exports at build time	—
Lazy loading	Defer offscreen images and components	—
Virtualization	Render only visible rows of a long list	—
Memoization	Skip recomputation and re-renders when inputs are unchanged	—
Debounce / throttle	Limit how often rapid events fire	—
Critical CSS	Inline the styles needed for the first paint	—

Delivery and offline

Term	Meaning
Bundler	Vite, esbuild, Webpack: turns source into deployable assets
Source map	Maps built output back to source; strip in production
PWA	Installable web app with a service worker and manifest
Service worker	Background script enabling offline and caching
Offline-first	Local store treated as primary, synced when connectivity returns
Optimistic sync	Accept local writes, reconcile with the server later

Part VIII — UI and UX design

UI is how it looks; UX is how it works and how it feels to use. The gap between amateur and professional interfaces is rarely aesthetic — it is the states nobody bothered to design.

Foundations

Term	Meaning	Prompt phrase
User flow	The path a user takes to complete a task	"Map signup to first action in under 4 steps"

Term	Meaning	Prompt phrase
Information architecture	How content and navigation are organised	"Group features into 5 top-level sections"
Wireframe	Low-fidelity layout with no styling	"Wireframe first, no colour"
Prototype	Clickable mock before real implementation	"Make it clickable before building"
Persona	A profile of the target user	"Design for a trader on mobile"
Affordance	A visual cue signalling what an element does	"Buttons must look clickable"
Cognitive load	Mental effort the interface demands	"One primary action per screen"
Progressive disclosure	Reveal advanced options only when needed	"Hide advanced settings behind a toggle"

Visual design

Term	Meaning	Prompt phrase
Visual hierarchy	Size, weight and contrast guiding the eye in order of importance	"Make the key metric the most prominent element"
Design system	Shared components and rules enforcing consistency	"Create a design system first"
Design tokens	Named values for colour, spacing, radius, typography	"Define tokens, no hardcoded hex values"
Typography scale	A consistent ratio-based set of sizes	"1.25 type scale, maximum 2 typefaces"
Spacing system	Consistent gaps on a 4 or 8 pixel grid	"8px spacing grid everywhere"
Contrast ratio	Text legibility against its background	"WCAG AA contrast minimum"
Whitespace	Empty space that creates clarity and grouping	"Generous whitespace, avoid clutter"
Colour semantics	Colour carrying consistent meaning	"Green for gain, red for loss, amber for warning only"

Term	Meaning	Prompt phrase
Dark mode	A second theme driven by the same tokens	"Support both themes via tokens"

Layout

Term	Meaning
Responsive design	Layout adapts to any viewport
Mobile-first	Design the smallest screen first, then expand
Breakpoints	Widths at which layout changes
Grid and flexbox	The two layout systems; grid for two dimensions
Above the fold	What is visible without scrolling
Touch target	Minimum 44 pixels so fingers can hit it

Interaction and feedback

Term	Meaning	Prompt phrase
Microinteraction	Small animated response to an action	"Subtle hover and press states"
Loading state	Skeleton or spinner while data arrives	"Skeleton loaders, never blank screens"
Empty state	What shows when there is no data yet	"Helpful empty state with a call to action"
Error state	Clear message plus a route to recovery	"Human-readable errors with a retry"
Optimistic UI	Update immediately, reconcile afterwards	"Optimistic update with rollback on failure"
Undo over confirm	Allow reversal instead of interrogating the user	"Undo toast for destructive actions"
Feedback	Every action is acknowledged	"Toast on success, inline on error"
Motion design	Purposeful animation, 150 to 300ms	"Animations under 300ms, respect reduced motion"

The four states every data-driven screen needs are loading, empty, error and populated. Designing only the last one is the most common gap in fast-built applications.

Usability principles

Principle	Meaning
Nielsen's heuristics	Ten classic usability rules; visibility of status and error prevention above all
Fitts's law	Larger, closer targets are faster to hit
Hick's law	More choices mean slower decisions
Jakob's law	Users expect your site to work like the sites they already know
Recognition over recall	Show options rather than requiring memory
Error prevention	Design so mistakes are difficult, not just recoverable

Principle	Meaning
Consistency	The same thing looks and behaves the same everywhere

Accessibility

Term	Meaning	Prompt phrase
WCAG 2.2 AA	The practical accessibility standard	"Meet WCAG 2.2 AA"
Semantic HTML	Correct elements carrying meaning	"Use semantic tags and alt text"
ARIA	Attributes adding meaning where HTML cannot	"ARIA only where semantics are insufficient"
Keyboard navigation	Fully operable without a mouse	"Visible focus states, logical tab order"
Colour independence	Meaning never carried by colour alone	"Add icons or labels beside colour coding"
Reduced motion	Respect the operating system setting	"Honour prefers-reduced-motion"
Screen reader	Software reading the interface aloud	"Test with a screen reader"

Research and validation

Method	What it gives
Usability testing	Five users surface most of the problems that affect roughly a third or more of users, given fixed realistic tasks. Rarer and subtler problems need far more participants
A/B testing	Which of two versions performs better on a chosen metric
Session replay and heatmaps	Where users click, hesitate and abandon
Task success rate	Measurable UX: did they finish, and how long did it take

Data-dense interfaces

Term	Prompt phrase
Information density	"Dense but scannable"
Data-ink ratio	"Minimal chart decoration"
Tabular numerals	"Monospaced digits, right-aligned numbers"
Freshness indicator	"Show last-updated time, flag stale data"
Destructive safeguards	"Confirm and summarise before irreversible actions"

Worth reading once: Refactoring UI, the Nielsen Norman Group articles, and lawsofux.com.

Part IX — Visual and special effects

One signature effect executed well beats ten competing for attention; restraint is what separates professional work from a template.

Surface and style

Effect	What it is
Glassmorphism	Translucent frosted panels using backdrop blur

Effect	What it is
Neumorphism	Soft extruded shapes from paired shadows; low contrast, use sparingly
Gradient and mesh gradient	Smooth multi-colour backgrounds
Elevation	A shadow system expressing depth layers
Glow	Light bloom marking active or focused elements
Grain and noise	Subtle texture overlay for a tactile feel
Depth of field	Blurring background content behind a modal

Interaction

Effect	What it is
Hover and press states	Lift, scale or colour shift on interaction
Ripple	Expanding circle from the click point
Magnetic button	Element drifts toward the cursor
3D tilt	Card rotates following pointer position
Spotlight	Radial highlight tracking the cursor
Spring physics	Momentum-based motion rather than linear easing

Motion and scroll

Effect	What it is
Fade, slide, scale	Basic enter and exit transitions
Stagger	Items animating in sequence rather than together
Skeleton shimmer	Moving highlight across loading placeholders
Shared element transition	One element morphing between views
Count-up and value flash	Numbers animating; cells flashing on change
Scroll-triggered reveal	Elements animating as they enter the viewport
Parallax	Layers moving at different speeds

Effect	What it is
Pinned section	Content stays fixed while the narrative scrolls
Scroll-driven sequence	Frames or 3D camera scrubbed by scroll position
Smooth inertia scroll	Weighted scrolling (Lenis)

Text and advanced

Effect	Tool
Split-text and scramble reveal	GSAP SplitText
Typewriter	CSS or JS
Gradient text and animated borders	CSS
Marquee ticker	CSS animation
Particle systems	tsParticles, Three.js
Shader backgrounds and distortion	GLSL with Three.js or OGL
3D scenes	Three.js, React Three Fiber
Post-processing (bloom, grain)	postprocessing library
Vector animation	Lottie, Rive
Confetti and celebration	canvas-confetti

Where effects belong

Project type	Use	Avoid
Landing page, portfolio	One shader or 3D hero, scroll reveals, split text	Stacking every effect
Dashboard, trading UI	Value flash, skeletons, smooth transitions	Particles, parallax, heavy shaders
Internal tools, ERP	Micro-feedback only	Anything decorative
Product showcase	Scroll-driven sequence	Heavy autoplaying scenes

Rules that keep effects professional

1. Every effect should communicate something: feedback, hierarchy or state. Decoration without meaning reads as noise.
2. Keep UI motion between 150 and 300 milliseconds; longer only for deliberate storytelling.
3. Animate only `transform` and `opacity`; animating width, height or position forces expensive layout work.
4. Honour `prefers-reduced-motion` without exception.
5. Lazy-load WebGL and 3D, cap device pixel ratio, and pause animation when the tab is hidden.
6. Test on a mid-range phone with CPU throttling; blur and particles destroy frame rates first there.
7. Content must be readable before the effect loads, and a static fallback must exist when WebGL is unsupported.

Part X — Reliability and resilience

Everything external will fail eventually, so resilience is the discipline of deciding in advance what happens when it does.

Term	Meaning	Prompt phrase
Timeout	Never wait indefinitely for an external call	"Set timeouts on every external call"
Retry with exponential backoff	Retry after growing delays: 1s, 2s, 4s	"Retry 3 times with backoff"
Jitter	Randomise retry delays so clients do not retry in lockstep	"Add jitter to backoff"
Thundering herd	Many clients retrying simultaneously and re-crushing a recovering service	What jitter prevents
Circuit breaker	After repeated failures, stop calling and fail fast; probe periodically	"Add a circuit breaker on the external API"
Bulkhead	Isolate resource pools so one failure cannot exhaust everything	"Separate connection pools per dependency"

Term	Meaning	Prompt phrase
Graceful degradation	Serve reduced function rather than failing entirely	"If the feed drops, show stale data with a warning"
Fail-safe default	On error or uncertainty, take the safe action	"On uncertainty, do nothing"
Fail-fast	Surface errors immediately rather than continuing in a bad state	"Validate and fail at the edge"
Backpressure	Slow or reject producers when consumers cannot keep up	"Bounded queue, reject when full"
Load shedding	Drop low-priority work to protect core function under overload	"Shed non-critical requests above 80% capacity"
Dead letter queue	Park repeatedly failing messages for inspection	"Route poison messages to a DLQ"
Idempotent retry	Retries are only safe when the operation is idempotent	The link between Parts III and VIII
Single point of failure	One component whose failure stops everything	"Identify every SPOF"
Redundancy	Duplicate components so one failure is survivable	"No single instance in the critical path"
Failover	Automatic switch to a standby instance	"Document the failover path"
Health check	Endpoints reporting liveness and readiness	"Add /health and /ready"
Graceful shutdown	Finish in-flight work before exiting	"Drain connections on SIGTERM"
Chaos engineering	Deliberately inject failures to verify resilience	"Write chaos tests for network drop and partial writes"
Blast radius	How much breaks when one thing breaks	"Limit the blast radius of this change"
Reconciliation	Periodically compare your state against the authoritative source	"Reconcile local state with the provider nightly"

Reliability targets

Term	Meaning
MTBF	Mean time between failures
MTTR	Mean time to recovery — usually the more useful number to improve
RTO	Recovery time objective: how long restoration may take
RPO	Recovery point objective: how much data you can afford to lose
Nines	99.9% is about 8.8 hours of downtime a year; 99.99% about 52 minutes

The discipline in one line: every external call needs a timeout, a bounded retry, and a defined behaviour when it finally fails.

Part XI — Performance, optimization and cost

Measure before optimizing: guessing the bottleneck is the most common way to waste a week.

The loop is profile, fix the single largest bottleneck, re-measure, repeat. Set a budget first ("p99 under 200 ms") and stop when you hit it.

Finding the bottleneck

Tool	Finds
cProfile, py-spy, scalene	Slow Python functions and CPU hot spots
memray, tracemalloc	Memory leaks and allocation bloat
EXPLAIN ANALYZE	Slow queries and bad plans
Locust, k6	The load at which throughput collapses
Browser DevTools, Lighthouse	Frontend load and render cost
Bundle visualizers	What is inflating the JavaScript bundle

Optimization by layer, in priority order

Layer	Typical fixes
Algorithm and design	Better data structures, removing repeated work, $O(n^2)$ to $O(n \log n)$
Database	Add indexes, fix N+1 queries, select fewer columns, batch writes, pool connections
I/O	Async for network calls, parallelize independent calls, batch requests
Caching	Redis or in-memory with TTL, memoize pure functions
Compute	Vectorize with NumPy or Polars, DuckDB for analytics, Numba or Rust for hot loops
Serialization	orjson over json, Parquet over CSV
Frontend	Code-splitting, lazy loading, image compression, memoization, list virtualization
Infrastructure	Compression, CDN, more workers, right-sized instances

Caching vocabulary

Term	Meaning
Cache hit ratio	Share of requests served from cache; the metric that matters
TTL	How long an entry stays valid
Cache invalidation	Removing stale entries — famously one of the hard problems
Cache stampede	Many requests recompute the same expired key at once
Write-through / write-behind	Update cache and store together, or store asynchronously
CDN	Edge servers serving static assets near the user

Term	Meaning
Edge caching	Caching responses at the CDN rather than the origin

Build and CI speed

Fix	How
Cache dependencies	Cache the package directory between CI runs
Parallel tests	pytest-xdist with `-n auto`
Test selection	Run only tests affected by changed paths
Layered Docker builds	Dependencies before source; multi-stage, slim base images
Faster tooling	uv over pip, ruff over flake8, esbuild or Vite over Webpack
Fail fast	Lint and unit tests first, slow suites last

Cost engineering

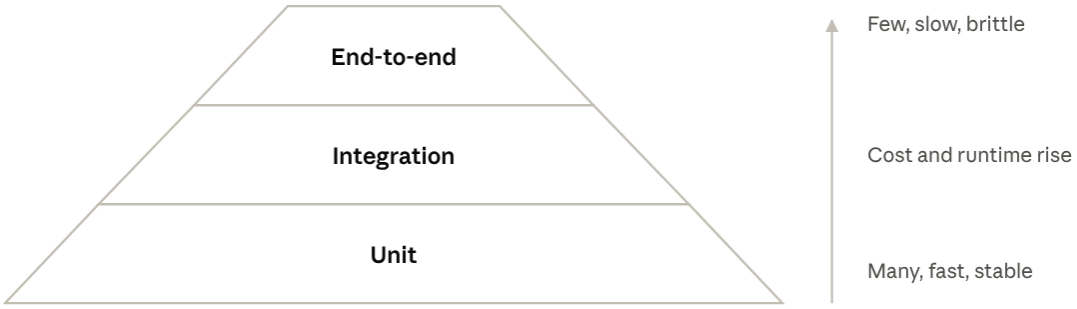
Term	Meaning
Cost per request	Infrastructure spend divided by traffic; the number to track
Egress charges	Data leaving the cloud provider, often the surprise line item
Storage tiering	Hot, warm and cold storage priced differently
Serverless vs provisioned	Serverless wins for spiky low volume, loses at steady high volume
Right-sizing	Matching instance size to measured usage
Idle cost	Resources billed while doing nothing

Two rules worth keeping: optimize correctness and clarity before speed, and make exactly one change per commit with before-and-after numbers attached.

Part XII — Testing and verification

You cannot test every combination — ten inputs with five values each is ten million cases — so the craft is covering the risk-weighted space deliberately and measuring what you covered.

Most tests should be fast unit tests



the testing pyramid · 3 tiers

Types of test

Type	Question it answers	Tool
Unit	Does each function do its job in isolation?	pytest, vitest
Integration	Do the components work together?	pytest with a test database
Contract	Does the API honour its schema for any input?	Schemathesis, Pact
End-to-end	Does a real user journey work?	Playwright, Cypress
Regression	Does old behaviour still hold?	Golden files, snapshots
Smoke test	Is the deployment basically alive?	A handful of critical paths
Load	Does it hold at expected traffic?	Locust, k6
Stress	Where exactly does it break?	Ramp until failure
Spike	Does it survive a sudden burst?	k6 spike profile
Soak	Does it leak or degrade over hours?	Long run with memory graphs
Chaos	Does it survive dropped networks and dead dependencies?	Toxiproxy, process kills

Type	Question it answers	Tool
Concurrency	Are there races or duplicates under parallelism?	Parallel clients, pytest-asyncio
Security	Injection, auth bypass, exposed data?	Bandit, OWASP ZAP

Taming the combinatorial explosion

Technique	What it does	Tool
Equivalence partitioning	Group inputs into classes, test one per class	Manual
Boundary value analysis	Test the edges: 0, 1, max, max+1, empty, null	Manual
Pairwise testing	Covers every pair of parameter values in ~20 cases instead of millions	PICT, allpairspy
Property-based testing	Define an invariant; the tool generates thousands of inputs to break it	Hypothesis
Fuzzing	Malformed, huge and hostile input	Atheris, Schemathesis
State-machine testing	Random valid sequences of operations	Hypothesis stateful
Mutation testing	Breaks your code deliberately; do your tests notice?	mutmut, Stryker
Coverage	Which lines and branches never ran	coverage.py

Mutation score is the honest metric. Ninety percent line coverage with weak assertions means nothing; mutation testing tells you whether the tests would actually catch a bug.

The condition matrix

Cross every feature against these axes to find the cases you would not have thought of.

Axis	Values to try
Data	Empty, one item, enormous, duplicate, malformed, unicode, negative, NaN

Axis	Values to try
Time	Clock skew, timezones, DST, expiry and cutoff boundaries
Network	Timeout, drop, slow, partial response, duplicate delivery
Dependencies	Database down, API returning 500, rate-limited, stale
Concurrency	One, ten and a thousand in parallel; the same key twice at once
Resources	Low memory, full disk, starved CPU
State	Cold start, restart mid-operation, crash recovery

Testing something already built

1. Write characterization tests that capture current behaviour before changing anything.
2. List the invariants in plain language: "position is never negative", "no duplicate orders", "debits equal credits".
3. Turn each invariant into a property-based test.
4. Fuzz the API edge against its schema.
5. Load test to establish a baseline, then find the breaking point.
6. Inject faults one at a time and confirm the system fails safe.
7. Run mutation testing to expose weak assertions.
8. Replay real historical data end to end, including the ugly days.
9. Automate all of it in CI so regressions cannot return.

Report results honestly: "stable to 500 requests per second, degrades at 800 due to database pool exhaustion" is worth more than "works under all conditions", which is never true.

Part XIII — Security and protecting your work

Security is layered, and broken access control — not exotic exploits — is the most common real-world breach.

Authentication and authorization

Term	Meaning
Authentication (AuthN)	Proving who you are
Authorization (AuthZ)	Deciding what you may do
Broken access control	Failing to check ownership per request; the top cause of breaches
Password hashing	bcrypt or argon2 with a salt; never store or reverse plaintext
MFA	A second factor beyond the password
Session management	HttpOnly, Secure, SameSite cookies; rotate on login; short expiry
JWT	Signed token carrying claims; verify signature and algorithm, keep it short-lived
Refresh token	Long-lived credential exchanged for new access tokens; revocable
OAuth 2.0 / OIDC	Delegated authorization, and identity layered on it
RBAC / ABAC	Access by role, or by attributes and context
Least privilege	Minimum permission needed, nothing more
Credential stuffing	Reusing breached passwords at scale; defended by rate limits and MFA

Input handling and injection

Threat	Defence
SQL injection	Parameterized queries; never concatenate SQL strings
XSS	Escape output, avoid raw HTML insertion, set a Content Security Policy
CSRF	SameSite cookies plus tokens on state-changing routes
Command injection	Never pass user input to a shell; use safe APIs

Threat	Defence
Path traversal	Normalize and allowlist file paths
SSRF	Allowlist outbound URLs, block internal IP ranges
Insecure deserialization	Never deserialize untrusted data with pickle or equivalents
Mass assignment	Explicit field allowlists on input models
XXE	Disable external entity resolution in XML parsers

Secrets, data and supply chain

Term	Meaning
Secrets management	Keys in environment variables or a vault, never in code
Secret scanning	gitleaks in pre-commit and CI; rotate anything ever committed
Key rotation	Scheduled replacement; revoke immediately on suspicion
Encryption at rest / in transit	Encrypted storage, and TLS on every hop including internal calls
PII redaction	Strip personal data, tokens and passwords from logs
Data minimization	Do not collect or retain what you do not need
Dependency audit	pip-audit, npm audit, Dependabot for known CVEs
Typosquatting	Malicious packages named like popular ones; verify before installing
SBOM	Software bill of materials: an inventory of what you ship
Container scanning	trivy on images; slim base, non-root user
Pinned CI actions	Pin third-party CI steps by commit SHA, not a moving tag

Infrastructure and response

Control	Why
Firewall and closed ports	Databases must never be publicly reachable
Environment isolation	Separate dev, staging and production credentials entirely
WAF and DDoS protection	Absorbs floods and blocks common attack patterns
Security headers	HSTS, CSP, X-Content-Type-Options
Audit log	Append-only record of sensitive actions: who, what, when
Alerting	Spikes in auth failures, 4xx or 5xx, or unusual volume
Incident plan	Written steps: contain, rotate keys, restore, notify
Canary token	Fake record or endpoint that alerts when touched

Reverse engineering, and defending against it

Understanding how software works without source is a standard security skill; the same knowledge tells you what you can and cannot protect.

Technique	What it does	Tool
Static analysis	Inspect code or binary without running it	Ghidra, IDA
Decompilation	Recover readable pseudo-source	Ghidra, JADX, dnSpy
Dynamic analysis	Observe behaviour while running	x64dbg, gdb, Frida
Traffic analysis	Inspect API calls and payloads	Wireshark, Burp Suite, mitmproxy
String extraction	Find hardcoded secrets and endpoints	`strings`
Patch diffing	Compare versions to locate a change	BinDiff
Frontend inspection	Unminify JavaScript, read Network and Sources tabs	Browser DevTools

Defence	Effect
Keep valuable logic server-side	The only real protection: there is nothing to reverse if it never ships
Minify and bundle	Raises effort for casual readers
Obfuscation	Slows analysis; friction, never a wall
Remove production source maps	Prevents instant recovery of readable source
Auth and rate limiting on every endpoint	Prevents abuse even when the logic is understood
Signed or encrypted payloads	Prevents tampering even when the structure is known
Watermarking	Traces a leak back to its source
Licence and terms	No technical barrier, but gives you legal recourse

The honest position: anything executing in the user's browser can be read by a determined person. Protection means architecture — proprietary logic runs on your server and the client receives only results.

Where to start

1. Ownership checks on every endpoint.
2. Parameterized queries and edge validation.
3. Secrets out of code; rotate anything already committed.
4. HTTPS, security headers, rate limiting.
5. Dependency and static analysis scanning in CI.
6. Logging and alerts on authentication failures.

Read the OWASP Top 10 once end to end; it is the standard checklist and takes an afternoon. The 2025 edition keeps broken access control at number one and folds SSRF into it, adds software supply chain failures at number three, adds mishandling of exceptional conditions at number ten, and moves injection down to number five.

Part XIV — Observability and operations

Observability is the ability to answer questions about your system that you did not anticipate when you built it; monitoring only answers the questions you did.

The three signals

Signal	Answers	Tools
Logs	What happened in this specific request	Structured JSON logs, Loki
Metrics	How the system behaves in aggregate over time	Prometheus, Grafana
Traces	Where the time went across services for one request	OpenTelemetry, Jaeger

Instrumentation vocabulary

Term	Meaning
Structured logging	Machine-parsable logs (JSON) rather than free text
Correlation ID	An ID attached to one request and propagated everywhere it travels
Span	One unit of work inside a trace, with a parent and a duration
OpenTelemetry	The vendor-neutral standard for emitting all three signals
Cardinality	Number of distinct label combinations; high cardinality destroys metric systems
Percentiles	p50, p95, p99 — averages hide the pain, percentiles reveal it
Golden signals	Latency, traffic, errors, saturation: the four to watch first
RED method	Rate, Errors, Duration — for request-driven services
USE method	Utilization, Saturation, Errors — for resources
Sampling	Keeping a subset of traces to control cost
Log level	DEBUG, INFO, WARN, ERROR used consistently and deliberately

Service levels

Term	Meaning
SLI	Service level indicator: the measured number, such as success rate
SLO	Service level objective: the target, such as 99.9% over 30 days
SLA	Service level agreement: the contractual promise with consequences
Error budget	The failure allowed by the SLO; when spent, stop shipping features and fix reliability
Burn rate	How fast the error budget is being consumed

Alerting and incidents

Term	Meaning
Actionable alert	Fires only when a human must act; everything else is a dashboard
Alert fatigue	Too many alerts, so real ones get ignored — the main failure of monitoring
Symptom-based alerting	Alert on user-visible symptoms, not internal causes
On-call rotation	Scheduled responsibility for responding
Escalation policy	What happens when nobody acknowledges
Runbook	Written steps for handling a known alert
Severity levels	SEV1 to SEV4: how much response an incident warrants
Blameless postmortem	Written analysis focused on systems and causes, never on individuals
Action items	Concrete follow-ups from a postmortem with owners and dates

The minimum worth having

Even a solo project deserves: structured JSON logs with a correlation ID, a health endpoint, error-rate and latency metrics, and one alert that reaches your phone when the error rate spikes. Everything else is refinement.

Part XV — Distributed systems

Once state lives on more than one machine, the hard problems are agreement and ordering, not throughput.

Agreement and coordination

Term	Meaning
Consensus	Getting nodes to agree on a value despite failures
Raft / Paxos	The two consensus algorithms in practice; Raft is the understandable one
Leader election	Choosing one node to coordinate writes
Quorum	A majority that must acknowledge before a decision counts (often W + R > N)
Split-brain	A partition leaves two nodes both believing they are leader
Fencing token	A monotonically increasing token that stops a stale leader from acting
Heartbeat	Periodic liveness signal; silence triggers failover
Gossip protocol	Nodes share state peer-to-peer rather than through a coordinator
Distributed lock	Mutual exclusion across machines; harder and more dangerous than it looks

Ordering and time

Term	Meaning
Clock skew	Machine clocks disagree; never trust wall-clock ordering across nodes

Term	Meaning
Logical clock	Counter-based ordering independent of wall time
Vector clock	Tracks causality across nodes, detects concurrent updates
Happens-before	The causal ordering relation underlying all of this
Idempotency (again)	The practical defence when ordering cannot be guaranteed

Messaging semantics

Term	Meaning
At-most-once	May be lost, never duplicated
At-least-once	Never lost, may be duplicated — the common default
Exactly-once	Marketing for at-least-once delivery plus idempotent processing
Message ordering	Guaranteed only within a partition or key, not globally
Consumer group	Parallel consumers splitting a stream's partitions
Offset / acknowledgement	Marks how far a consumer has processed

Transaction patterns across services

Pattern	What it solves
Two-phase commit (2PC)	Atomic commit across systems; blocking and rarely used at scale
Saga	A long transaction as a chain of local transactions with compensating undo steps
Outbox pattern	Write the event to a table in the same transaction as the data, then publish it — removes the dual-write problem
CQRS	Separate write model from read model, each optimised for its job

Pattern	What it solves
Event sourcing	Store the sequence of events as truth; current state is a replay of them
Idempotent consumer	Deduplicate by message ID so redelivery is safe

The honest summary: distributed systems are not faster systems, they are systems that survive partial failure. Adopt them when availability demands it, not for speed.

Part XVI — Pipelines, CI/CD and infrastructure

A pipeline is a chain of automated stages where each stage's output feeds the next, and the discipline is that every stage is idempotent, validated and observable.

Kinds of pipeline

Pipeline	What flows through	Typical stages
Continuous integration	Code	Lint, build, test, scan
Continuous delivery	A built artifact	Package, stage, deploy, verify, roll back
Data (ETL / ELT)	Raw data	Extract, transform, load
Streaming	Live events	Ingest, process, sink
ML	Data to model	Features, train, evaluate, deploy, monitor
Security (DevSecOps)	Code and dependencies	Static analysis, secret scan, dependency audit

Pipeline vocabulary

Term	Meaning
Stage / step	One unit of work in the chain
Artifact	The output of a stage: a build, a model, a dataset
DAG	Stages and dependencies as a directed graph with no cycles
Orchestrator	Schedules and runs the graph: Airflow, Prefect, Dagster

Term	Meaning
Trigger	What starts a run: a push, a schedule, an event
Idempotent stage	Re-running produces the same result, which is what makes retries safe
Checkpointing	Saving progress so a failure resumes rather than restarts
Backfill	Re-running over historical periods
Data lineage	Tracing where each output came from
Data validation	Schema and quality checks between stages (Pandera, Great Expectations)
Quality gate	A check that blocks the pipeline when it fails
Fan-out / fan-in	Splitting work in parallel, then joining results

Containers and infrastructure

Term	Meaning
Container	Application plus dependencies in an isolated, portable image
Image layer	Cached build step; order the Dockerfile so dependencies cache
Multi-stage build	Build in one image, copy only the result into a slim runtime image
Orchestration	Scheduling containers across machines (Kubernetes, ECS)
Pod / service / ingress	Kubernetes units: running group, stable address, external entry
Infrastructure as code	Infrastructure defined in version-controlled files (Terraform, Pulumi)
Immutable infrastructure	Replace servers rather than mutating them
Configuration drift	Reality diverging from the declared state
Environment parity	Dev, staging and production kept as similar as possible

Term	Meaning
Service mesh	Infrastructure layer handling service-to-service traffic

Deployment strategies

Strategy	How it works
Rolling deploy	Replace instances gradually
Blue-green	Two environments; switch traffic at once, roll back instantly
Canary	Send a small traffic share to the new version, watch metrics, then widen
Feature flag	Ship code dark, enable it separately from deployment
Shadow traffic	Mirror real requests to the new version without serving its responses
Rollback	Return to the previous version; must be one command

Continuity

Term	Meaning
Backup and restore	Automated, encrypted, and tested — an untested backup is not a backup
Disaster recovery	Plan to restore service after major failure, measured by RTO and RPO
Multi-region	Running in more than one geography for latency or survival
Runbook	Step-by-step instructions for a known operational task

A minimal pipeline worth having from day one: push triggers lint, then unit tests, then a security scan, then a container build, then deploy to staging, then smoke tests, then a manual approval, then production.

Part XVII — Engineering process and product thinking

What separates a senior engineer from a productive one is the ability to work in a way other people can build on.

Version control and collaboration

Term	Meaning
Atomic commit	One logical change per commit, with a message explaining why
Conventional commits	Structured messages (`feat:` , `fix:`) enabling automated changelogs
Trunk-based development	Short-lived branches merged daily into main; fewer merge conflicts
Git flow	Long-lived release branches; heavier, suited to versioned products
Pull request	A proposed change opened for review
Code review	Reading for correctness, clarity and risk, not style preferences
Rebase vs merge	Rewrite history for a linear log, or preserve it as it happened
Cherry-pick	Apply one commit onto another branch
Bisect	Binary search through history to find the commit that broke something
Semantic versioning	MAJOR.MINOR.PATCH: breaking, additive, fix
Changelog	Human-readable record of what changed between versions
Monorepo vs polyrepo	One repository for many projects, or one each

Documentation that earns its keep

Artifact	Purpose
README	What it is, how to run it, how to test it — the first thing anyone reads

Artifact	Purpose
ADR	Architecture decision record: context, options, decision, consequences
Runbook	Steps for a known operational task or alert
API documentation	Generated from the schema, never hand-maintained
C4 model	Four zoom levels: context, container, component, code
Postmortem	Blameless analysis of an incident with action items

An ADR is the highest-value document in AI-assisted work: it records why a choice was made, which is exactly the context a future session cannot reconstruct from code alone.

Code quality

Term	Meaning
Technical debt	Deliberate or accidental shortcuts that raise the cost of future change
Refactoring	Changing structure without changing behaviour, under test
Code smell	A surface symptom of a deeper design problem
Cyclomatic complexity	Number of paths through a function; high values resist testing
DRY	Do not repeat yourself — but duplication is cheaper than the wrong abstraction
YAGNI	You are not going to need it; build for today's requirement
SOLID	Five object-oriented design principles
Linting and formatting	Automated style enforcement so review discusses substance
Type checking	Static types catching errors before runtime (mypy, TypeScript)
Pre-commit hooks	Checks that run before a commit is accepted

Product thinking

Term	Meaning
MVP	The smallest version that tests the core assumption
User story	"As a [user], I want [goal], so that [benefit]"
Acceptance criteria	Conditions that define done, written before building
Scope creep	Requirements expanding mid-build without re-planning
Prioritization	RICE or MoSCoW: deciding what not to build
North star metric	The one number that best reflects delivered value
Funnel	Stage-by-stage conversion with drop-off between stages
Retention and churn	Who keeps using it, and who leaves
Activation	The first moment a user gets real value
Instrumentation	Events logged to answer product questions later
Dogfooding	Using your own product daily

The product question worth asking before writing any code: what would make me abandon this approach? If there is no answer, the plan is a wish rather than a plan.

Part XVIII — Domain notes: trading, ML and LLM systems

Three domains where the general principles acquire specific, non-obvious requirements.

Trading and financial systems

Term	Meaning
Order lifecycle	A state machine: new, pending, partially filled, filled, cancelled, rejected
Idempotent order placement	A client-generated order ID preventing duplicates on retry

Term	Meaning
Kill switch	A single control halting all automated activity immediately
Risk gate	Server-side limits checked before every order leaves
Position reconciliation	Periodic comparison of local state against the broker's record
Slippage	Difference between expected and executed price
Latency budget	Allowed time per stage from tick to order
Market microstructure	How orders, spreads and liquidity actually interact
Backtest	Simulating a strategy over historical data
Look-ahead bias	Accidentally using information unavailable at that moment
Survivorship bias	Testing only on instruments that still exist today
Overfitting	A strategy tuned to historical noise rather than signal
Walk-forward validation	Repeated train-then-test on rolling windows
Paper trading	Live data, simulated execution
Audit trail	Immutable record of every decision and order for later inspection

The architectural rule for this domain: strategy logic runs server-side only, every write is idempotent, and nothing trades without a risk gate and a kill switch.

Machine learning systems

Term	Meaning
Feature store	Central, versioned store of features shared by training and serving
Training-serving skew	Features computed differently in production than in training
Data drift	Input distribution changes over time

Term	Meaning
Concept drift	The relationship between inputs and target changes
Data leakage	Target information contaminating the features
Train/validation/test split	Three disjoint sets; the test set is touched once
Cross-validation	Repeated splits for a more reliable estimate
Class imbalance	Rare positives making accuracy a misleading metric
Precision and recall	False positives versus missed positives; choose by cost
Model registry	Versioned models with lineage and metrics
Shadow deployment	New model scored alongside the old without acting on it
Model monitoring	Watching live performance and drift after deployment
Reproducibility	Pinned data, code, seeds and environment

LLM and AI applications

Term	Meaning
RAG	Retrieval-augmented generation: fetch relevant documents, then generate from them
Embedding	Vector representation capturing semantic meaning
Vector database	Store supporting nearest-neighbour similarity search
Chunking	Splitting documents for retrieval; chunk size dominates RAG quality
Reranking	A second pass reordering retrieved results by relevance
Context window	How much text the model can consider at once
Prompt injection	Untrusted input hijacking instructions; treat retrieved content as data, never commands

Term	Meaning
Grounding	Requiring answers to cite retrieved sources
Hallucination	Fluent, confident, false output
Evals	Automated test suites scoring output quality against expectations
Guardrails	Validation of model output before it is used or displayed
Token cost control	Caching, truncation and model routing to manage spend
Agent and tool use	A model calling functions; every tool needs its own authorization

The discipline that matters most here is evals: without a scored test set, every prompt change is a guess, and quality drifts invisibly.

Part XIX — The system design interview

Interviewers assess how you reason under ambiguity, so the structure of your answer matters more than the architecture you land on.

The framework

Step	Minutes	What to do
1 Clarify requirements	5	Functional needs, then non-functional: scale, latency, consistency, availability. Ask before assuming
2 Define scope	2	State explicitly what you will not design
3 Estimate capacity	5	Users, requests per second, storage, bandwidth
4 Design the API	5	A handful of endpoints with request and response shapes
5 Design the data model	5	Entities, relationships, access patterns, choice of store
6 Draw the high-level design	10	Client, load balancer, services, cache, database, queue

Step	Minutes	What to do
7 Scale it	10	Where it breaks first, then sharding, replication, caching, CDN
8 Handle failure	5	Each component dying, and the resulting degradation
9 Discuss tradeoffs	3	What you chose, what you rejected, and under what conditions you would choose differently

Numbers worth memorizing

Operation	Order of magnitude
L1 cache reference	1 nanosecond
Main memory reference	100 nanoseconds
Read 1 MB sequentially from memory	100 to 250 microseconds
SSD random read	100 to 150 microseconds
Round trip within one datacentre	0.5 milliseconds
Disk seek, spinning disk	10 milliseconds
Round trip across continents	150 milliseconds
Seconds in a day	86,400, usefully rounded to 100,000
1 million writes a day	about 12 per second
A typical database row	100 bytes to 1 KB

Worked estimation pattern

For a service with 10 million daily active users each making 20 requests:

- 200 million requests a day, divided by about 100,000 seconds, gives roughly 2,000 requests per second average.
- Peak traffic is commonly 2 to 3 times average, so design for about 5,000 per second.
- If each request stores 1 KB, that is 200 GB a day, about 73 TB a year before replication.
- A read-heavy ratio of 100 to 1 means caching the hot 20% removes most database load.

The method matters more than precision: state your assumption, round aggressively, and show the arithmetic.

What interviewers reward

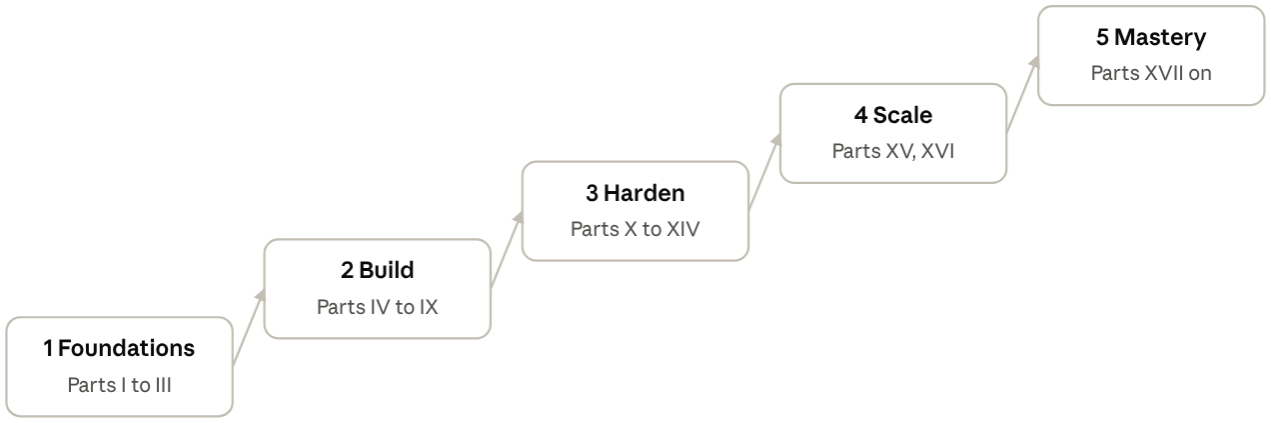
Behaviour	Why
Asking clarifying questions first	Shows you do not build on assumptions
Starting simple, then scaling	Premature complexity is a negative signal
Naming tradeoffs out loud	The core skill being tested
Identifying the bottleneck before fixing it	Demonstrates engineering judgement
Admitting uncertainty	Confident wrongness is worse than "I would measure that"
Driving the conversation	Silence reads as being stuck

Classic problems to practise

URL shortener, rate limiter, news feed, chat system, notification service, file storage, web crawler, ride-hailing dispatch, ticket booking with no double-selling, real-time price feed. Work through five of these properly rather than skimming twenty.

Part XX — Study path and prompt templates

Study in dependency order — the parts of this encyclopedia are arranged to follow it, so reading front to back is the path. Each stage assumes the one before it, and skipping ahead produces vocabulary without judgement.



the study path · 5 stages

The five stages

Stage	Read	Build to prove it	You have finished when
1 Foundations	Parts I to III	A small CRUD API with a real database and a sensible schema	You can define every term in Part I in one sentence and name the tradeoff it sits on
2 Build	Parts IV to IX	One real project with a rules file, specs and a usable interface	You ship a feature end to end without the AI drifting off-spec
3 Harden	Parts X to XIV	Add tests, logging, error handling and a security pass to that project	You can state the measured limits of your own system
4 Scale	Parts XV and XVI	Add caching, a queue, a CI/CD pipeline and a load test	You can explain where your system breaks first and why
5 Mastery	Parts XVII to XIX, XXI to XXIII	Take one project to production quality, then practise design problems	You can defend every architectural decision you made

The common failure is spending months in stage 1 reading and never entering stage 2. Move to building as soon as you can define the terms; depth comes from debugging, not from reading.

How long each stage takes

Roughly two to four weeks per stage if you build alongside reading, and indefinitely if you only read. Stage 5 is not an endpoint but a practice you repeat on every project.

Prompt template library

Starting a project

> Act as a principal engineer. Before any code, write a spec: goals, non-goals, data model, API surface, failure modes, and the constraints I have given. List every assumption you are making and ask me about the ones that matter. Wait for my approval.

Choosing an architecture

> Give me three architectures for this, with tradeoffs on complexity, cost, latency and operational burden. Recommend one and state the conditions under which that recommendation would be wrong.

Implementing a feature

> Modular monolith. Define schemas first. Separate routes, services and data access. Idempotent writes, transactions, timeouts and bounded retries on external calls, structured logs with correlation IDs, config in YAML, unit tests with mocked externals. Fail safe on error. Change only the files I name and show me the diff.

Reviewing what was built

> Act as a hostile reviewer who did not write this code. Find race conditions, data-loss paths, unhandled failures and security holes. Rank findings by severity with file and line. Do not fix anything yet.

Testing

> Audit this codebase for testability. List the invariants and the riskiest modules. Then write characterization tests, property-based tests for each invariant, a load profile and fault-injection tests. Run everything and report real output, including coverage and mutation score. Mark anything you could not run as UNVERIFIED.

Optimizing

> Profile this project. Report the top five bottlenecks with measured numbers. For each, propose a fix with expected gain and risk. Apply one at a time, re-measure after each, and keep the tests passing.

Security pass

> Audit against the OWASP Top 10. For each finding give severity, location, an exploit scenario and a fix. Check auth and per-request ownership, injection, secrets in code and git history, CORS, rate limiting, dependency CVEs and sensitive data in logs.

Before shipping

> Assume this failed in production three months from now. List the ten likeliest causes and the mitigation for each. Then tell me which of those mitigations are missing today.

The habits that compound

1. Keep a rules file and update it whenever you correct the same mistake twice.
2. Read every diff. Code you cannot explain is code you cannot defend in an interview or debug at 2am.
3. Own the hard twenty percent yourself: the data model, the core logic, the security boundary.
4. Measure and report honestly. The anti-fabrication instinct is rarer than technical skill and more valuable.
5. Finish one project properly rather than starting ten. Depth is the only thing that distinguishes you now that everyone can generate code.

Part XXI — Concurrency, boundaries and safe change

Six subjects that the earlier parts assume but never teach: how work runs at the same time, where to draw service boundaries, and how to change a running system without stopping it.

Concurrency models

Race conditions appeared in Part III as a symptom. This is the machinery underneath them.

Term	Meaning
Concurrency vs parallelism	Dealing with many things at once, versus doing many things at once
Process vs thread	Separate memory and isolation, versus shared memory and cheap switching
Thread pool	Fixed set of workers consuming a queue of tasks
Event loop	One thread interleaving many waiting tasks; the model behind async/await and Node
Blocking vs non-blocking	Whether a call holds its thread while waiting on I/O
CPU-bound vs I/O-bound	Processes for the first, async or threads for the second
Global interpreter lock	In the default CPython build, one thread executes bytecode at a time, so threads help I/O but not CPU work. Python 3.14 ships an officially supported free-threaded build without the GIL, but it is not the default and must be installed and run separately
Mutex / lock	Mutual exclusion over shared state
Semaphore	Permits limiting how many run at once
Atomic operation	Completes indivisibly; no interleaving is observable
Critical section	The code that must not run concurrently
Deadlock / livelock / starvation	Mutual waiting, busy non-progress, and never getting a turn
Lock contention	Threads queuing on the same lock; often worse than the work itself
Lock-free structures	Concurrency via atomic compare-and-swap rather than locks
Immutability	Data that cannot change needs no lock — the cheapest concurrency strategy
Actor model	Isolated actors owning state, communicating only by message (Erlang, Akka)
CSP	Independent processes communicating over channels (Go)

Term	Meaning
Structured concurrency	Tasks scoped to a block, so none outlive or escape it

The practical rule: prefer immutability and message passing over shared mutable state. Most concurrency bugs are shared state that someone forgot to protect, and they surface under load rather than in tests.

Domain-driven design

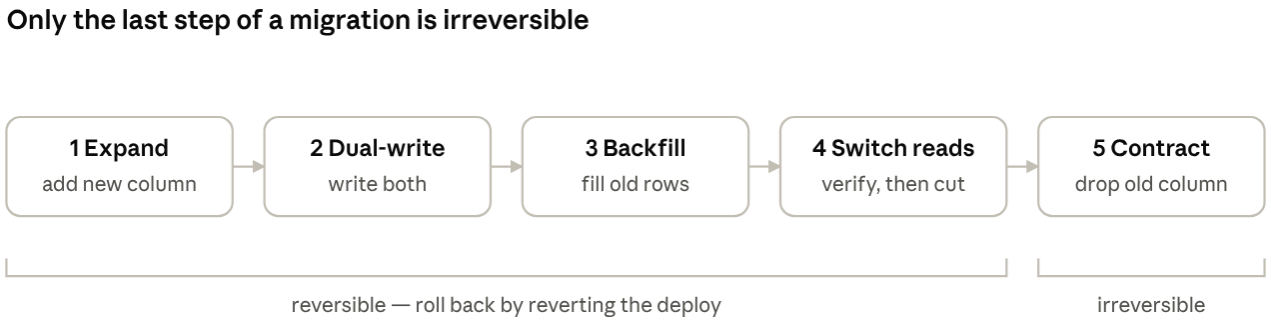
This is the vocabulary for answering "where should the boundary go", which Part II raised but did not resolve.

Term	Meaning
Ubiquitous language	One shared vocabulary used by code, tests and conversation alike
Bounded context	A boundary inside which a term has exactly one meaning
Context map	How bounded contexts relate and what they exchange
Entity	Has identity that persists through change (a user, an order)
Value object	Defined entirely by its values and is immutable (money, a date range)
Aggregate	A cluster of objects changed as one unit, with a consistency boundary
Aggregate root	The only entry point into an aggregate
Repository	Collection-like interface for loading and saving aggregates
Domain event	Something meaningful that happened, named in the past tense
Anti-corruption layer	Translation layer stopping another system's model leaking into yours

The useful heuristic: a service boundary should coincide with a bounded context and an aggregate's consistency boundary. If two things must change in the same transaction, they belong in the same service.

Zero-downtime schema change

Part III covered migrations; this is how to run one against live traffic without a maintenance window.



expand-contract migration · 5 phases

Term	Meaning
Expand-contract	Add the new shape, migrate, then remove the old — never rename in place
Dual-write	Write both old and new representations during transition
Backfill	Populate the new column or table for rows that already exist
Shadow read	Read both and compare, without serving the new one yet
Online schema change	Tools performing the change without long table locks (gh-ost, pt-osc)
Lock-free migration	Avoiding operations that block writes on large tables
Reversibility window	The span during which rollback is still possible

Every phase here ships as its own deploy. The rule that makes it safe: old and new code must both work against the schema at every intermediate step.

Conway's law and team topologies

Term	Meaning
Conway's law	A system's architecture mirrors the communication structure of the organisation that built it
Inverse Conway manoeuvre	Reshape teams deliberately to get the architecture you want
Stream-aligned team	Owns one product stream end to end
Platform team	Builds internal tooling other teams consume self-service
Cognitive load	The amount of system one team can hold in its head; the real limit on service size
Ownership boundary	Who gets paged when it breaks — often the most honest definition of a service

This explains most microservice decisions better than any technical argument: services usually split along team lines, not along load.

Multi-tenancy

Model	Isolation	Cost
Shared schema, tenant ID column	Weakest; a missing filter leaks data across customers	Cheapest
Schema per tenant	Moderate; migrations multiply	Moderate
Database per tenant	Strongest; operationally heavy	Most expensive

Term	Meaning
Tenant isolation	Guaranteeing one customer can never read another's data
Row-level security	Database-enforced tenant filtering, independent of application code
Noisy neighbour	One tenant's load degrading everyone else

Term	Meaning
Per-tenant quotas	Rate and resource limits applied per customer
Tenant-aware caching	Cache keys must include the tenant, or you leak across customers

The failure mode to fear: a query that forgets its tenant filter. Enforce it in the database or a shared repository layer, never by remembering to add a `WHERE` clause.

Little's law and queueing

Term	Meaning
Little's law	Concurrency = arrival rate × average latency
Utilization	Share of capacity in use; queue length grows sharply above about 70–80%
Queue depth	Work waiting; a rising queue means arrival exceeds service rate
Head-of-line blocking	One slow item delaying everything behind it
Tail latency amplification	Fanning out to many services makes the slowest one dominate p99
Coordinated omission	Load tests that stop sending during a stall, hiding the worst latency

A worked use: 500 requests per second at 200 ms average latency needs 500 × 0.2 = 100 concurrent slots. If your pool holds 50, requests queue and latency rises, which raises concurrency further — the feedback loop behind most sudden collapses.

Part XXII — Practices and common subsystems

The remaining practices that recur in professional work, and the handful of subsystems almost every product ends up building.

Serialization and schema evolution

Term	Meaning
Serialization format	JSON (human-readable), Protobuf and Avro (binary, typed, compact), MessagePack

Term	Meaning
Schema registry	Central store of message schemas with compatibility enforcement
Backward compatible	New code reads old data
Forward compatible	Old code reads new data; the harder of the two
Full compatibility	Both directions, required for independent deploys
Safe change	Adding an optional field with a default
Unsafe change	Removing a field, renaming one, changing a type, or reusing a field number

The rule that prevents most incidents: never reuse a removed field's name or tag number. Reserve it permanently.

Error handling philosophy

Term	Meaning
Exceptions vs result types	Thrown errors versus errors returned as values the caller must handle
Error taxonomy	A deliberate classification: validation, not-found, conflict, transient, internal
Retryable vs terminal	Whether retrying could ever succeed; the single most useful distinction
Error budget for retries	Caps so retries cannot amplify an outage
Fail loudly	Never swallow an exception silently; log with context or re-raise
Error wrapping	Adding context while preserving the original cause
User-facing vs internal errors	What the user reads versus what the log records; never leak stack traces
Panic vs recover	Unrecoverable versus handled failure, as separate paths

Coordination patterns

Term	Meaning
Orchestration	A central coordinator tells each service what to do; easy to follow, single point of change
Choreography	Services react to events independently; loosely coupled, harder to trace
Workflow engine	Durable execution of long-running processes (Temporal, Step Functions)
Compensating action	The undo step for an already-committed stage
Timeout and escalation	What happens when a step never completes

Rate limiting algorithms

Algorithm	Behaviour
Fixed window	Count per interval; allows a double burst at the boundary
Sliding window log	Exact, memory-hungry
Sliding window counter	Weighted approximation; the usual production choice
Token bucket	Tokens refill at a steady rate; permits controlled bursts
Leaky bucket	Constant drain rate; smooths output completely
Concurrency limit	Caps simultaneous in-flight requests rather than rate

Testing in production

Term	Meaning
Synthetic monitoring	Scripted journeys run continuously against production
Real user monitoring	Measuring actual user sessions in the field

Term	Meaning
Automated canary analysis	Statistically comparing canary metrics against baseline, auto-rolling back
Dark launch	Running new code without exposing its output
Game day	A scheduled rehearsal of a failure scenario

Compliance and data governance

Term	Meaning
GDPR / DPDP	European and Indian data protection regimes; consent, purpose limitation, user rights. India's DPDP Rules were notified in November 2025 with phased enforcement, most obligations landing 18 months on
Right to erasure	Users can demand deletion — which breaks naive event sourcing and backups
Data residency	Legal requirement that data stays within a jurisdiction
Retention policy	How long each data class is kept before mandatory deletion
Audit requirements	Who accessed what, when, retained immutably
PII and PCI scope	Which systems touch personal or card data; minimising scope minimises obligation
Data processing agreement	Contract governing a vendor's handling of your users' data

Open-source licensing

Licence	Obligation
MIT, BSD, Apache 2.0	Permissive; attribution. Apache also grants patent rights
LGPL	Linking is fine; modifications to the library must be shared

Licence	Obligation
GPL	Derivative works must be released under GPL — consequential for commercial products
AGPL	GPL extended to network use, so running it as a service triggers the obligation
SSPL, BSL	Source-available, not open source; commercial restrictions apply
Unlicensed code	No licence means no permission to use it, whatever the repository implies

This is routinely ignored and occasionally expensive. Check the licence of anything that reaches production, including code an AI reproduced from training data.

Developer experience

Term	Meaning
Time to first commit	How long a new machine takes to run the project and ship a change
One-command setup	A single script bringing up the whole stack
Devcontainer	Containerised development environment, identical for everyone
Seed data	Realistic sample data so local work is possible without production access
Hot reload	Code changes visible without restart
Local-first development	Everything runnable offline, nothing requiring shared staging

Common subsystems and their known patterns

Subsystem	What it actually involves
Search	Inverted index, tokenisation and stemming, relevance scoring (BM25), faceting, typo tolerance, index freshness

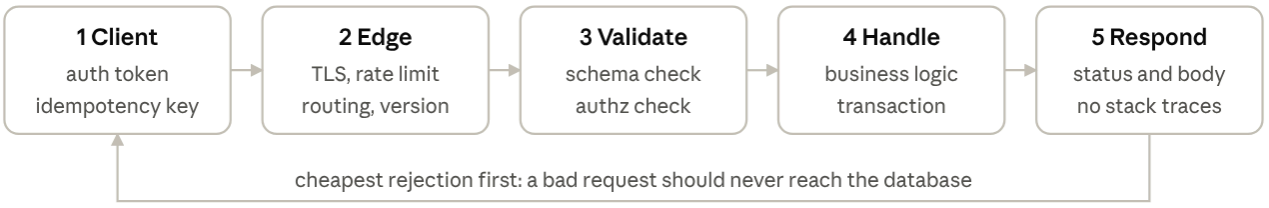
Subsystem	What it actually involves
Payments	Idempotent charges, webhook handling, reconciliation against the provider, refunds and chargebacks, PCI scope minimisation, never storing card data
Notifications	Channel routing, user preferences, batching and digests, deliverability and bounce handling, quiet hours, deduplication
File and media	Direct-to-storage signed URLs, virus scanning, transcoding pipelines, CDN delivery, thumbnail generation
Background jobs	Durable queue, retries with backoff, idempotent handlers, scheduled and recurring work, visibility into failures
Feature flags	Targeting rules, gradual rollout percentages, kill switches, flag cleanup as technical debt
Audit logging	Append-only, tamper-evident, who-what-when-where, queryable, retained to policy
Import and export	Streaming large files, partial failure reporting, validation before commit, resumability

Each of these looks trivial until built. Payments and notifications in particular are where teams discover that idempotency, retries and reconciliation were not optional.

Part XXIII — APIs from zero to hero

An API is a contract: a promise about what you accept, what you return, and what happens when something goes wrong. Everything below is about making that promise precise and keeping it.

A request should be rejected at the earliest stage that can reject it



the request path · 5 stages

Level 0 — The anatomy of a request

Every HTTP exchange has the same five pieces. Knowing them by name makes every later level readable.

Piece	What it carries	Example
Method	The intended action	`GET`, `POST`, `PUT`, `PATCH`, `DELETE`
URL and path	Which resource	`/v1/orders/42`
Query string	Filtering, sorting, pagination	`?status=open&limit=50`
Headers	Metadata about the request	`Authorization`, `Content-Type`, `Idempotency-Key`
Body	The payload, usually JSON	`{"symbol": "INFY", "qty": 10}`

The response mirrors it: a status code, headers, and a body.

Level 1 — Methods and status codes

Method	Purpose	Idempotent	Has body
GET	Read a resource	Yes	No
POST	Create, or trigger an action	No	Yes
PUT	Replace a resource entirely	Yes	Yes
PATCH	Update part of a resource	Not inherently	Yes

Method	Purpose	Idempotent	Has body
DELETE	Remove a resource	Yes	No

Code	Meaning	Use when
200 OK	Success with a body	Reads and updates
201 Created	Resource created	Include a `Location` header
202 Accepted	Accepted for async processing	Long-running work
204 No Content	Success, nothing to return	Deletes
400 Bad Request	Malformed or invalid input	Validation failure
401 Unauthorized	Not authenticated	Missing or invalid credentials
403 Forbidden	Authenticated but not permitted	Ownership or role failure
404 Not Found	No such resource	Also used to hide existence from unauthorised users
409 Conflict	State conflict	Duplicate, or version mismatch
422 Unprocessable	Syntactically valid, semantically wrong	Business rule violation
429 Too Many Requests	Rate limited	Include `Retry-After`
500 Internal Server Error	Your fault, unexpected	Never leak the stack trace
503 Service Unavailable	Temporarily down or overloaded	Include `Retry-After`

The commonest beginner mistake is returning 200 with an error message in the body. Clients cannot handle that generically, and every retry, cache and monitor downstream reads it as success.

Level 2 — Designing resources

Rule	Good	Bad
Nouns, not verbs	`POST /orders`	`POST /createOrder`

Rule	Good	Bad
Plural collections	`/orders/42`	`/order/42`
Nest only one level	`/orders/42/items`	`/users/1/orders/42/items/7/tax`
Filter with query params	`/orders?status=open`	`/orders/open`
Actions as sub-resources	`POST /orders/42/cancel`	`POST /cancelOrder`
Consistent casing	`snake_case` everywhere	Mixed styles across endpoints
Opaque IDs	`/orders/ord_8f2a`	`/orders/42` exposes volume and invites enumeration

Level 3 — Correctness

This is where an amateur API becomes a reliable one.

Concern	What to do
Validation	Validate at the edge with a typed schema. Reject unknown fields rather than ignoring them
Idempotency	Accept an `Idempotency-Key` on POST; store the outcome against it and replay the stored response on repeat
Pagination	Cursor-based for anything that changes; return `next_cursor`, never a total count on large tables
Sorting and filtering	Allowlist sortable fields; arbitrary field names invite injection
Partial responses	Let clients request fewer fields rather than building parallel endpoints
Consistent errors	One error shape across every endpoint
Time and money	ISO 8601 with timezone; integer minor units for currency, never floats
Bulk operations	Define partial-failure semantics explicitly before building them

A single error shape, used everywhere:

```
{
  "error": {
    "code": "insufficient_funds",
    "message": "Order exceeds available margin.",
    "details": [{"field": "qty", "issue": "max 8 at current margin"}],
    "request_id": "req_01HX3"
  }
}
```

The `code` is for machines and never changes; the `message` is for humans and may. The `request_id` is what a user quotes to support and what you grep for in logs.

Level 4 — Security

Concern	What to do
Authentication	Short-lived bearer tokens; refresh tokens stored and revocable
Authorization	Check ownership on every single request, never only in the UI
Object-level checks	`/orders/42` must verify 42 belongs to the caller — the top API vulnerability
Rate limiting	Per key and per IP; return 429 with `Retry-After`
Input size limits	Cap body size, array lengths and nesting depth
CORS	Explicit origin allowlist; never a wildcard with credentials
HTTPS only	With HSTS; reject plaintext outright
Secrets	Keys in headers, never in query strings, which end up in logs and referrers
Mass assignment	Bind only allowlisted fields; never pass the request body straight into a model
Error leakage	Generic messages externally, full detail in logs

Level 5 — Evolution

Term	Meaning
URL versioning	`/v1/orders`; crude, obvious, and the most common
Header versioning	Version negotiated via a header; cleaner, harder to debug
Additive change	New optional fields are safe to ship any time
Breaking change	Removing or renaming a field, tightening validation, changing a type or status code
Deprecation policy	Announce, set a sunset date, emit a `Deprecation` header, then remove
Tolerant reader	Clients ignore unknown fields, so you can add them freely
Contract test	Automated check that the API still satisfies its published schema

Treat a published API as permanent. Shipping v2 means operating v1 and v2 together for as long as clients exist.

Level 6 — Performance

Technique	Effect
ETag and `If-None-Match`	Returns 304 and no body when nothing changed
`Cache-Control`	Lets clients and CDNs skip the request entirely
Compression	gzip or brotli on responses
Avoiding N+1	Fetch related records in one query, not one per row
Projection	Return only the fields needed, not whole objects
Connection reuse	HTTP/2 multiplexing and keep-alive
Async for slow work	Return 202 with a status URL instead of holding the connection
Payload limits	Cap page size so one client cannot request everything

Level 7 — Operating it

Practice	Why
OpenAPI specification	Generates documentation, clients and contract tests from one source
Generated documentation	Hand-written docs drift; generated ones cannot
Request IDs	Correlate a user's complaint with your logs
Per-endpoint metrics	Rate, errors and duration broken down by route
SLOs	Published targets for latency and availability
Changelog	Clients need to know what changed and when
Sandbox environment	Lets integrators build without touching production
Client SDKs	Generated from the spec, so they cannot drift

Level 8 — Beyond REST

Style	Choose when
REST over HTTP	Public APIs, broad client compatibility, cacheability
gRPC	Internal service-to-service calls where latency and typed contracts matter
GraphQL	Many client shapes querying a rich graph; accept the caching and rate-limiting complexity
Webhooks	You must push events to someone else; sign payloads, expect retries, make handlers idempotent
Server-sent events	One-way live updates to browsers
WebSocket	Genuinely bidirectional, persistent interaction

The checklist before publishing an endpoint

1. Correct method and status codes, with no 200-wrapped errors.
2. Typed schema validation at the edge, rejecting unknown fields.
3. Ownership and permission checked on this specific resource.
4. Idempotency key honoured on every non-idempotent write.

5. Pagination on every collection, with a capped page size.
6. The standard error shape, carrying a request ID.
7. Rate limits, body size limits and timeouts in place.
8. OpenAPI entry, with at least one contract test.
9. Metrics and structured logs emitting for this route.
10. Documented as additive-only, with a deprecation path if it ever changes.

An endpoint satisfying all ten is indistinguishable from one shipped by a mature engineering team. That is the whole of the distance from zero to hero.

Part XXIV — One system, designed end to end

Every part before this teaches vocabulary in isolation. This one designs a single system from requirements to failure modes, using the interview framework from Part XIX, and names the part each decision comes from.

The problem: an order-placement service that accepts trade orders from users, applies risk checks, and submits them to an external broker. It is chosen because it exercises idempotency, transactions, queueing, external failure and auditability — most of this book.

1. Requirements

Functional	Non-functional
Place, cancel and query an order	A duplicate order is never created, under any retry
Apply risk limits before submission	p99 under 300 ms from request to accepted
Reflect broker fills back to the user	Every decision auditable after the fact
Show current positions	Available during market hours above all else

The hard requirement is the second one. It dictates almost every later decision.

2. Scope

In scope: order lifecycle, risk gating, broker submission, status. Out of scope: the strategy that decides what to trade, settlement, tax reporting, and the user interface. Stating this matters — Part XIX rewards it, and it stops the design sprawling.

3. Capacity estimate

Assumptions, stated so they can be challenged: 50,000 active users, 20 orders each per trading day.

- 50,000 × 20 = **1 million orders a day**.
- Trading concentrates at the open. Assume 30% of volume in the first 30 minutes: 300,000 ÷ 1,800 seconds ≈ **167 orders per second average at peak**.
- Allow 3x for bursts: design for **500 orders per second**.
- At 1 KB per order record, 1 GB a day, about **365 GB a year** before replication.
- Status reads outnumber writes roughly 100:1, so peak reads approach **17,000 per second** — which is a caching problem, not a database problem.
- Little's law (Part XXI): 500 per second at 50 ms service time needs 500 × 0.05 = **25 concurrent slots**. A pool of 25 is the floor, not the target.

4. The API

Following Part XXIII.

Endpoint	Notes
`POST /v1/orders`	Requires `Idempotency-Key`. Returns 201 with the order, or replays the stored response
`GET /v1/orders/{id}`	Cacheable with ETag; ownership checked per request
`GET /v1/orders?cursor=`	Cursor pagination, capped page size
`POST /v1/orders/{id}/cancel`	Action as a sub-resource; idempotent by nature
`GET /v1/positions`	Served from cache, with a freshness timestamp

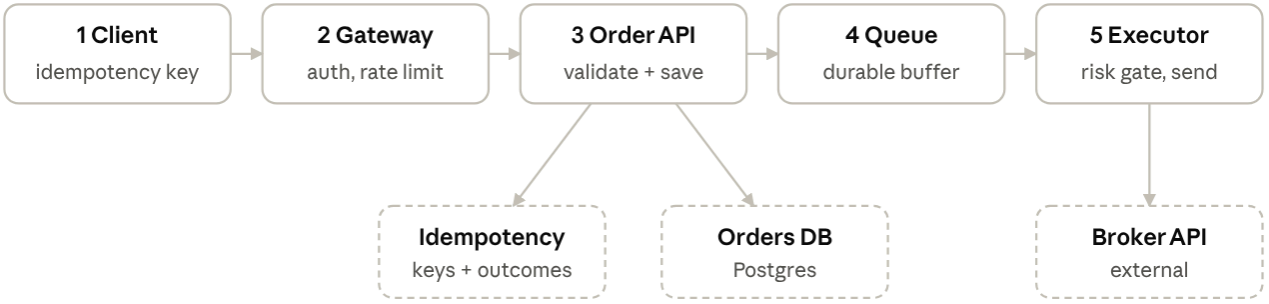
5. Data model

Table	Key columns	Why
`orders`	id, user_id, symbol, qty, side, status, version	`version` enables optimistic locking (Part III)
`idempotency_keys`	key, user_id, request_hash, response, expires_at	Replays the stored outcome; `request_hash` catches key reuse with different parameters

Table	Key columns	Why
`order_events`	order_id, seq, event, payload, created_at	Append-only audit trail (Part XVIII)
`positions`	user_id, symbol, qty	Derived; reconciled against the broker, never trusted blindly

6. Architecture

Every write is made idempotent before it reaches the broker



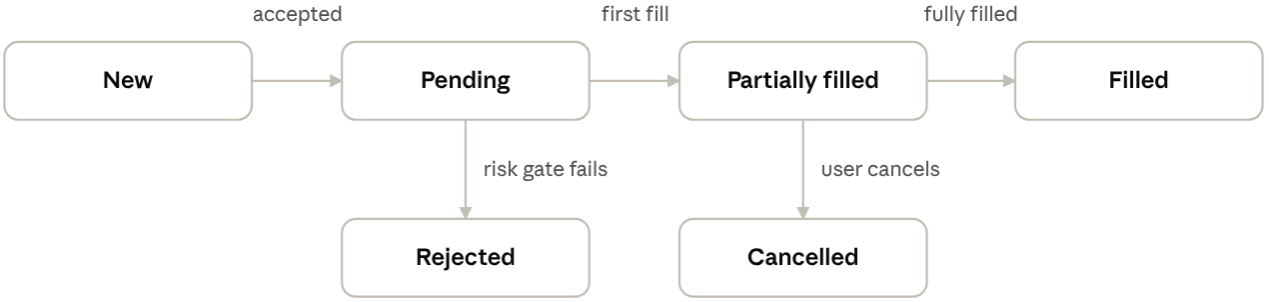
order service architecture · 5 services, 3 stores

The write path in order: the gateway authenticates and rate-limits; the Order API validates the schema and checks the idempotency key; a single transaction writes the order row and its first event; the order is published to a durable queue; the executor applies risk limits and calls the broker.

The decision that carries the most weight: the API acknowledges the order **once it is durably persisted**, not once the broker confirms. Holding the connection open across an external call would tie your availability to the broker's — Part X's argument against it.

7. The state machine

Every order ends in filled, cancelled or rejected



order state machine · 6 states

States are explicit and transitions are the only way to move between them. Every transition appends to `order_events`, so the audit trail is a by-product of the design rather than extra work.

8. Where it breaks first

Pressure	First failure	Fix
Market-open burst	Connection pool exhausted, latency climbs, concurrency climbs further	Bounded queue with backpressure (Part X); shed status reads before writes
Status polling	Database read load, not write load	Cache positions and order status with short TTL (Part XI)
Broker slow or down	Executor threads block, queue grows	Timeout, circuit breaker, orders stay queued and visibly pending (Part X)
Broker returns duplicates	Position drift	Client order ID on every submission; nightly reconciliation (Part XVIII)
One user floods the API	Everyone degrades	Per-user rate limits and quotas (Part XXI, multi-tenancy)
Growth past one database	Writes contend	Shard by user_id — deferred until measured, since it costs cross-user queries (Part II)

9. Tradeoffs made

Chosen	Rejected	Why
Modular monolith	Microservices	One team; Conway's law says services would split along team lines that do not exist yet (Part XXI)
PostgreSQL with an event log	Full event sourcing	Append-only events for audit, but current state kept as rows — simpler to query, and the right-to-erasure problem is avoided (Part XXII)
Async submission	Synchronous broker call	Keeps availability independent of the broker

Chosen	Rejected	Why
Strong consistency on orders	Eventual consistency	Money. Part I's CAP tradeoff resolved toward C
Cursor pagination	Offset pagination	Order lists change constantly under the reader

10. What would make this wrong

The design assumes retail-scale latency tolerance. If the requirement were sub-millisecond execution, almost every choice above inverts: no queue, no network hop to a separate executor, in-process risk checks, and a colocated deployment. Naming the condition under which your design fails is the step Part XIX rewards most and candidates skip most.

Working through it yourself

Redesign this for one changed requirement at a time and see what moves: the broker guarantees nothing about duplicates; orders must survive a full datacentre loss; one user generates 90% of volume; regulators require seven-year retention. Each one touches a different part of this book, which is the point.

Verification note

This encyclopedia was compiled in a single session and then spot-checked against external sources on 3 November 2026. Most of it has not been independently verified, and that limitation is stated here rather than left for the reader to discover.

Checked and confirmed correct

Claim	Source
Core Web Vitals thresholds: LCP under 2.5s, INP under 200ms, CLS under 0.1, at the 75th percentile	corewebvitals.io
WCAG 2.2 is the current W3C Recommendation (October 2023); WCAG 3.0 remains a draft	W3C / Wikipedia

Checked and corrected

Was	Now	Source
"In CPython, one thread executes bytecode at a time" stated without qualification	Qualified: true of the default build, but Python 3.14 ships a supported free-threaded build without the GIL	The Register
Read 1 MB sequentially from memory: 10 microseconds	100 to 250 microseconds	Latency numbers gist, updated figures
SSD random read: 20 microseconds	100 to 150 microseconds	As above
"Five users surface roughly 80% of problems"	Qualified: most problems affecting a third or more of users, under fixed realistic tasks	MeasuringU
DPDP named without timeline	Rules notified November 2025, phased enforcement over 18 months	Exterro
OWASP Top 10 referenced generically	2025 edition changes noted	Semgrep

Not verified

Everything else. That includes the bulk of the definitions in every part, all named tools and libraries, the UI/UX and effects guidance, and the trading, ML and LLM material in Part XVIII. Treat a specific number or tool name here as a starting point to confirm, not as a citation.

One structural limitation also stands: there are no graded exercises, so nothing here tests whether you have learned it. Part XXIV now works one system end to end and closes with variations to attempt, which is the nearest this book comes to practice. Treat it as a reference with one worked example, not a course.